

SECV3263 MULTIMEDIA WEB PROGRAMMING

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ALTERNATIVE ASSESSMENT – INDIVIDUAL SELF REFLECTION

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Role: Animation / interaction Designer

For this project, I was responsible for developing the Skills Showcase section of our Interactive 3D Portfolio Website using Three.js. My main contributions included building the 3D skill spheres, member tabs, and skill panel, as well as implementing hover and click interactions and keyboard shortcuts to improve user navigation. In addition, I developed the glowing aura particle effect around each skill orb by implementing the makeSmokeTexture function and a particle system, which enhanced the visual appeal and creativity of the Skills Showcase section.

Throughout the development process, I gained valuable experience in using Three.js to create interactive 3D web components and particle effects. One of the main challenges I faced was ensuring that the particle system performed smoothly while maintaining responsive interactions with the skill spheres. I also encountered difficulties in coordinating the animations and interactions without affecting the overall website performance. To overcome these challenges, I experimented with different particle settings, optimized the rendering process, debugged the interaction logic, and referred to the Three.js documentation to achieve a balance between visual quality and performance.

Overall, this project strengthened my technical skills in JavaScript and Three.js while improving my problem-solving and teamwork abilities. I learned the importance of optimizing interactive 3D elements to provide a smooth user experience without compromising performance. If I were to enhance this project in the future, I would explore more advanced particle effects, smoother animation transitions, and additional interactive features to make the Skills Showcase even more engaging. This experience has increased my confidence in developing creative and interactive multimedia web applications using modern web technologies.